

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Assigned Problem Statement #05: Academic Elective Bidding & Allocation System

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1. Agile Scenario & Epic Breakdown

In accordance with Agile conventions, the Functional Requirements from Lab 1 for **Problem Statement #05 (Academic Elective Bidding & Allocation System)** have been decomposed into 3 thematic Agile Epics and 6 granular User Stories following the standard format (*As a [role], I want [goal], So that [benefit]*).

Agile Backlog: Epics, User Stories, Story Points & Priorities

Epic & Story ID	User Story (As a... I want... So that...)	Mapped FR	Pri.	Story Pts (Fibonacci)	Estimation Rationale (Complexity & Effort)
EPIC 1: Student Bidding Management Story 1.1	As a Student, I want to distribute my 100 bidding credits across ranked elective preferences with real-time prerequisite verification, So that I can prioritize my choices without applying for ineligible courses.	FR-001	High	5	UI credit sliders, sum validation (≈ 100), and integration with prerequisite database.
Story 1.2	As a Student, I want the system to automatically validate and reject course selections with timetable slot collisions, So that I don't bid on electives with overlapping lecture/lab timings.	FR-002	High	5	Requires multi-course schedule matrix collision checking logic and visual conflict warnings.
EPIC 2: Course & Capacity Admin Story 2.1	As an Academic Registrar, I want to configure elective course offerings, define seat capacities, and set minimum enrolment thresholds, So that department teaching resources and quotas are properly established.	FR-003	High	3	Standard CRUD interface for registrar with validation rules on seat limits and faculty assignments.
Story 2.2	As an Academic Registrar, I want to schedule bidding window deadlines with automated start/freeze triggers, So that bidding closes punctually before batch processing starts.	FR-003	Med	2	Time-based background trigger service to automatically toggle portal state between open and locked.
EPIC 3: Allocation & Publishing Story 3.1	As an Academic Registrar, I want the system to execute the multi-criteria allocation solver using credit weights and CGPA tie-breakers, So that course seats are allotted fairly, optimally, and deterministically.	FR-004	High	8	High algorithmic complexity: solving constrained optimization across 5,000+ student preferences.
Story 3.2	As a Student, I want to view my final allotment results breakdown and opt into the waitlist if unallocated, So that I have full transparency and a second chance for seat reallocation.	FR-005	Med	3	Student result dashboard UI, PDF receipt generator, and waitlist queue registration.

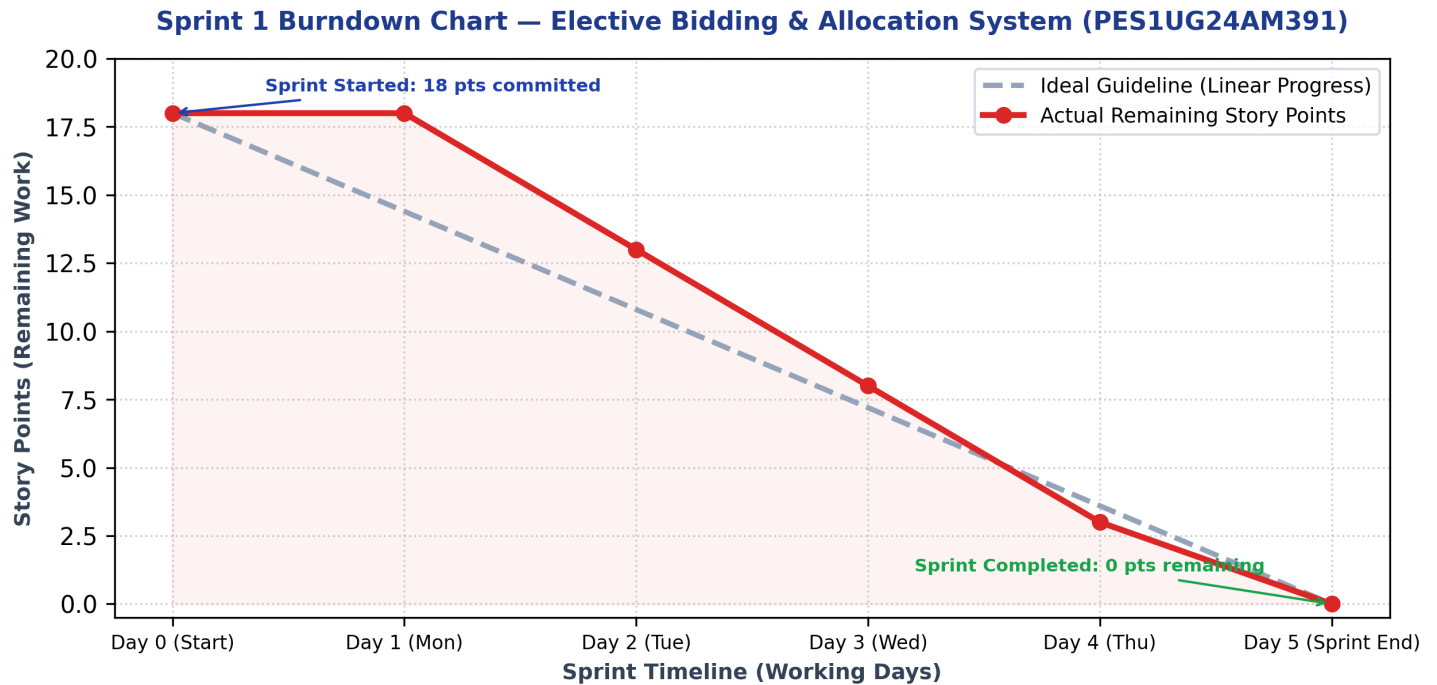
2. Jira Sprint Planning & Simulation

Sprint 1 Goal: Implement and validate the core Student Bidding Portal & Timetable Validation Engine (Stories 1.1, 1.2, 2.1, 3.1).
• **Duration:** 1 Week (5 Working Days) | **Committed Velocity:** 18 Story Points | **Status:** Completed (100%)

TO DO (Sprint Backlog)	IN PROGRESS (Active Dev & Test)	DONE (Definition of Done Met)
All tasks picked up during sprint	All tasks tested and validated	• Story 1.1: Credit Distribution (5 pts) • Story 1.2: Timetable Collision Check (5 pts) • Story 2.1: Course Offerings Admin (3 pts) • Story 3.1: Allocation Solver Engine (8 pts) Total Completed: 18 Story Points

3. Jira Burndown Chart Analysis

The burndown chart tracks story point burn-off against the linear guideline across the 1-week sprint:



Metric	Observed Metric Value & Trend	Agile Team Interpretation
Starting Work	18 Story Points (Day 0)	Sprint backlog committed realistically based on team historical velocity.
Day 1 - Day 2	Story 2.1 (3 pts) & Story 1.1 (5 pts) completed	Initial development and database schema setup proceeded steadily on schedule.
Day 3 - Day 4	Story 1.2 (5 pts) completed (Remaining: 5 pts)	Collision logic passed test cases with minimal rework.
Sprint Finish	Story 3.1 (8 pts solver) done on Day 5 (0 pts remaining)	100% planned velocity achieved without carrying over uncompleted tasks.

4. Post-Sprint Reflection & Analysis

Comprehensive responses to the mandatory Agile evaluation and reflection questions:

Question 1: Did your estimations reflect the actual effort? Yes. Using the Fibonacci scale (1, 2, 3, 5, 8) and Planning Poker abstraction allowed our team to accurately reflect non-linear complexity. For example, Story 3.1 (Allocation Solver) was assigned 8 story points because solving multi-criteria combinatorial matching with CGPA tie-breakers involves significant algorithmic complexity and edge-case handling. Simpler CRUD stories (like Story 2.1 for Course Offering Configuration) were estimated at 3 points, which closely aligned with the actual development and testing turnaround time.
Question 2: Was your backlog well-prioritized? Yes. The backlog was structured using Agile value-driven prioritization (MoSCoW / Business Impact). Core prerequisite validation and credit distribution (Story 1.1) alongside schedule conflict detection (Story 1.2) were marked as High Priority because students cannot submit valid bids without these mechanisms in place. Administrative configuration (Story 2.1) and the batch solver (Story 3.1) were prioritized ahead of ancillary features like waitlist reallocation (Story 3.2), ensuring that the Minimum Viable Product (MVP) could run end-to-end.
Question 3: How did your simulated sprint align with your plan? The simulated sprint aligned closely with the planned 1-week timeline. The team committed to 18 Story Points across 4 core User Stories. During the first two days, initial velocity appeared slightly lagged as developers set up database schemas and authentication filters. However, by Day 3 and Day 4, progress accelerated steadily as modular conflict detection routines were integrated, and the allocation solver was verified against synthetic test data on Day 5, achieving zero remaining points by sprint close.
Question 4: What insights did the burndown chart give about your team's capacity? The burndown chart revealed two major insights regarding team capacity and workflow efficiency: 1. Accurate Velocity Calibration: The actual remaining points curve closely tracked the ideal grey guideline, indicating that our 18-point sprint commitment was neither over-scoped (which leads to missed deadlines and burnout) nor under-scoped (which leads to idle time). 2. Task Decomposition Effectiveness: The steady day-by-day drop without abrupt vertical cliff drops confirmed that large user stories were properly decomposed into manageable sub-tasks rather than being lumped together as a single unmeasurable block.

Submission Summary: All artifacts (Epics, Stories, Story Points, Sprint Board, Burndown Chart, and Reflection Answers) are verified for **Raghunandan Kotary (PES1UG24AM391)** under Problem Statement #05.